DESIGN AND VERIFICATION DOSSIER

Is the tube in the stomach

Nasogastric aspirate pH bench prototype

COURSE	EXSU 620 Surgical Innovation • Project 04
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Abstract

We developed a bench prototype that measures nasogastric aspirate simulant pH, applies the course decision rule and saves an audit record. The revised browser service can queue phone messages for all four completed outcomes, including technical failures.

Three added prototype–meter comparisons differed by +0.082, +0.105 and +0.231 pH, meeting the selected ±0.30 agreement criterion. Reference uncertainty, repeatability and accuracy near the decision thresholds remain unverified.

Two historical phone receipts have separate consent and confirmation records. The revised notification path passed targeted software checks. Phone receipt for each outcome under the revised service were demonstrated with photo evidence.

Comparator uncertainty, physical fault coverage, complete alarm-pattern verification and seven-day measurement stability remain open, and only tested against a demo environment.

All measurements concern educational bench liquids. Course verdict wording is reproduced to document the prototype and does not authorize a clinical feeding decision.

Guide to read the report

What the prototype does and what the evidence means

We designed the prototype to measure pH, check reading quality and save results. In this report, the host is the laptop and its Python monitor service, which communicates with the Pico, runs the browser interface, logs results and produces audio. The operator enters initials and a sample name, immerses the probe and selects NEW SAMPLE. The Pico takes 20 scheduled readings and publishes a result or technical refusal at approximately 21 seconds. The result is held until STOP or NEW SAMPLE.

Green, amber and red represent the course's pH categories. Cyan represents a measurement problem. Laptop speakers provide sound because the separate amplifier overheated during development. The physical button stops the test, silences laptop audio when its event reaches the service, and commands the ring off. The browser's ACKNOWLEDGE control mutes sound but does not itself send a stop command to the Pico.

The archived final logs show timed acquisition, advisory and critical results, stopped samples and a confirmed phone message. Added snapshots include a valid normal result. The videos show normal/critical colours, critical tones and physical STOP/OFF with audio cessation. A supplied image shows the disconnected technical display. Software tests exercise boundaries, age and defined faults; they do not establish true reference pH or detection of every damaged electrode.

The prototype agreed with the comparison meter within ±0.30 pH at three tested points. Earlier checks against nominal buffer labels exceeded this criterion. The reference changed between these comparisons; the new results do not demonstrate that the earlier discrepancy was corrected. Reference uncertainty, repeatability and accuracy near the decision thresholds remain unverified.

The critical alarm is remembered by the laptop separately from the current result. The ring follows the current sample and can turn off after stopping while critical history remains. This is a declared difference from a continuously latched red indication. Laptop or service failure also removes audio and does not restore alarm history automatically on restart.

The optional phone message reports the completed sample outcome or the reason the system withheld a verdict. The operator checks the matching sample ID on the phone and records receipt separately.

How to read the evidence references: C1 denotes the course material; E01–E23 identify project files, recordings or analyses listed in Sources and reproducibility. These are internal cross-references, not publication numbers. For example, E13 is the monitor software, E19 the light/alarm video and E21 the physical ACK/STOP video. Log line numbers refer to records in the named JSONL file, not pages of this report.

I. Clinical need and requirements (Deliverable D1)

Need statement and intended scope

A way to reduce uncertainty during nasogastric tube placement checks in patients requiring enteral feeding in order to support timely, appropriate feeding decisions.

The prototype aims for numerical pH measurement, explicit decision categories, refusal when the measurement cannot be trusted, and a retrievable record. Our prototype explores this workflow using bench liquids. No patient samples, tube-location ground truth or clinical outcomes are included. The intended benefit of reduced uncertainty has therefore not been measured in clinical practice. [C1]

Clinical requirements, as issued

ID	REQUIREMENT
CR1	The pH of a nasogastric aspirate shall be measured and displayed to one decimal place, with a clear indication of when the reading has settled.
CR2	The device shall state a verdict against the accepted rule: at or below pH 5.5 gastric, 5.6 to 6.0 escalate, above 6.0 do not feed.
CR3	The device shall refuse to give a verdict when its calibration is older than the configured interval, or when the electrode is drifting or unresponsive.
CR4	Every test shall be recorded with the value, the verdict, the operator and the calibration state, and shall be retrievable for audit.

Engineering Requirements

The following eight engineering requirements retain the draft's numerical targets and identify the latest implemented settings. These requirements reflect our final design, including changes we made during development. Not all of them were established before testing began. Requirements are assessed separately from whether an individual run completed. [E01-E04]

ID AND TRACE	REQUIREMENT AND ACCEPTANCE CRITERION
ER1 • CR1	Estimate pH over the proposed nominal range 4.00-10.01 and display one decimal. Every independent reference check must be within ±0.30 pH, with reference identity, temperature and uncertainty documented. A display or a two-point fit alone does not pass accuracy.
ER2 • CR1 CR3	Acquire readings at seconds 1-20 after accepted sample start, each within 250 ms of schedule. Publish no verdict before 21.000 s and one outcome by 21.250 s on the device clock. A valid outcome also requires a ten-reading pH range ≤0.35 maintained for a ten-second dwell.
ER3 • CR2	Apply categories to the unrounded final filtered estimate. pH ≤5.5 is NORMAL; 5.5 < pH ≤6.0 is ADVISORY; pH >6.0 is CRITICAL. Values immediately below, at and above both boundaries must classify correctly. Invalid measurements must not produce a verdict.
ER4 • CR3	Before a verdict, reject incomplete or late captures, implausible input and the defined drift or response-check failure. Any raw value <500 or >60000, estimated pH outside 0-14, drift >0.40 pH, missing timed reading or failed known-buffer response check must produce a technical result with no verdict at the decision point.
ER5 • CR3 CR4	Limit calibration use to 604800 seconds from recorded completion. A restart or new sample must not renew the interval. Missing clock, invalid coefficients, future completion time and the expiry boundary must withhold a new verdict. Physical accuracy throughout that interval requires separate verification.
ER6 • CR2 CR3	Provide the specified normal, advisory, critical and technical indications; retain critical history until acknowledged and a valid normal returns. Normal is green and silent, advisory amber with three tones, critical red with ten, and technical cyan with one. The physical button must stop acquisition, mute sound through the host and command OFF. Current indication and retained critical history must be distinguishable.
ER7 • CR3 CR4	Attribute each accepted NEW SAMPLE to a distinct sample ID, operator, label and calibration context. Reset its timer and acquisition history. After link loss or more than five seconds without a valid heartbeat, withhold the host's current verdict. Reconnection must require a new sample. Corresponding JSONL and SQLite events must agree, and interruptions and label corrections must remain retrievable.
ER8 • CR2 CR4	With prior consent, queue one message per eligible completed Normal, Advisory, Critical or Technical sample. Include the sample ID and outcome, with the reason for technical refusal. Match the provider message ID to an independently observed phone receipt. Idle, measuring and manually stopped cases shall produce no result message.

The evaluated nominal range is broader than the 4-7 calibration interval. Values above pH 7 are extrapolated. ER1 is a target to be verified, not an established continuous operating range. ER2 measures software acquisition and publication timing; it does not guarantee electrode equilibrium within 20 seconds.

The signal limits in ER4 are provisional software settings. They are neither measured fault boundaries nor electrical protection. ADC level of pH 10.01 was documented as 45000; the latest code uses 60000 to allow measuring up to pH 14.00. The 14.00 was not involved in the project therefore the ADC reading does not exist. It is one of the limitations. [E01, E03, E08]

The drift rule compares the mean of readings 1-3 with readings 18-20. A difference exactly equal to 0.40 pH passes; a greater absolute difference fails in routine mode. This is a net-change test. It can respond to a jump and may miss changes that return to their initial level.

Stuck assessment is limited to a dedicated known-buffer response check. A three-reading moving average must remain within ±0.10 pH of the previous measured baseline throughout seconds 3-20 for at least 15 seconds to fail. The actual observation interval is approximately 17 seconds. Routine NEW SAMPLE measurements do not assess stuck behaviour. [E03]

ER6 uses a deliberate deviation from continuously latched red indication: the ring shows the current sample and physical STOP turns it off, while the host retains critical history. Approval of this latch deviation is not documented. The separately reported instructor approval for laptop audio does not establish approval of latch behaviour. Selected functions are demonstrated; remaining physical pattern coverage is stated in D7.

Audit and notification requirements

Acceptance rationale

A seven-day calibration policy was chosen to reduce repeated calibration work. It is not a manufacturer-validated lifetime established by this project. A stored timestamp permits age checking, while a longitudinal reference study is needed to justify the duration physically. No new sample, browser action or reboot should be described as recalibration.

The latest stability condition is a maximum-minus-minimum range of 0.35 pH across ten unfiltered readings, followed by a ten-second qualifying dwell. The evidence notes describe ±0.35 around a mean for 20 seconds; that is a different algorithm and is not used here. The calibration script's separate 0.95 pH capture-spread limit is also not an accuracy tolerance. [E01, E03, E04]

Classification uses full internal precision before the packet is rounded. Consequently, the final recorded value 6.029 is critical even though the browser displays 6.0. This avoids silently rounding a value down across the decision boundary, but the display still needs to explain such cases to users. [E06]

CLINICAL REQUIREMENT	ENGINEERING REQUIREMENTS	MAIN EVIDENCE
CR1	ER1 ER2	Calibration record, reference checks, timed captures and display code. Partial. Three paired checks met the numerical tolerance. Independent reference documentation and full range remain open.
CR2	ER3 ER6 ER8	Boundary tests, final results, alarm records and phone receipts. Partial. Tested rules pass. Reconcile one-decimal boundaries and latched red behaviour with the card. Revised phone receipt remains open.
CR3	ER2 ER4 ER5 ER6 ER7	Defined virtual faults, calibration-age tests, stop and interruption records. Partial. Selected refusals pass. Routine stuck detection and physical fault coverage remain open.
CR4	ER5 ER7 ER8	Calibration identity, paired stores, sample attribution and consent records. Supported for reviewed runs. Current stores reconcile. Crash recovery and revised notification receipts remain open.

II. System architecture (Deliverable D2)

The sensing chain consists of a glass pH electrode, BNC connection and pH meter V1.1 conditioning board. The board converts the electrode response into an analogue signal read at GP26 on the Pico 2 W through the Grove shield. The Pico also controls the 16-pixel ring on GP15 and reads an active-low button on GP14. [E03, E17]

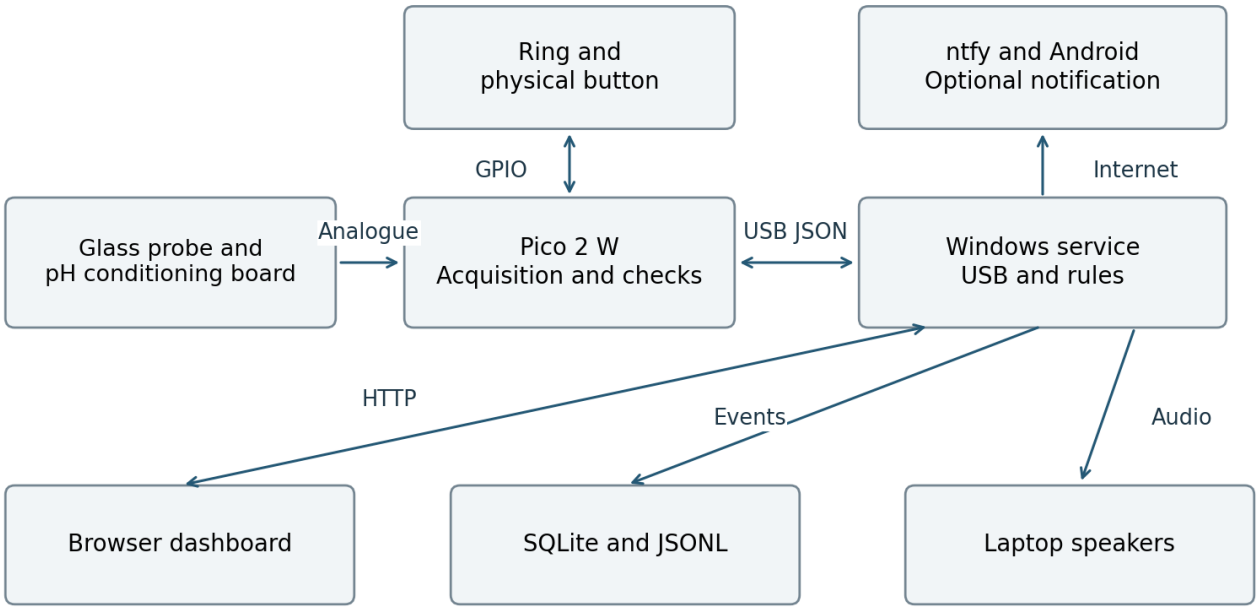


Figure 1. Implemented functional architecture. The Pico performs measurement checks. The Windows service owns serial communication, audit writes and audio. The browser is an interface to that service. Remote notification is a separate internet-dependent path. [E03, E13]

The Pico uses MicroPython rather than the CircuitPython runtime suggested in the project card. It transmits newline-delimited JSON over USB. Its wireless capability is not used for measurement transport. The service exposes a local HTTP interface at 127.0.0.1:8765 and supplies a browser display with a trace, result, calibration details and recent events.

USB provides both power and communication in the documented arrangement. Laptop audio replaces the separate PAM8302 amplifier and speaker following overheating during assembly. The failed amplifier is not counted as a working subsystem. [E01, E02]

The architecture is dependent on a functioning laptop and service. Local logging and display do not require internet, but phone messages do. The prototype has not demonstrated independent audible warning when the laptop loses power or the service stops.

Calibration and measurement processing

The calibration script asks for nominal pH 4.00 and 7.00 liquids. It waits 60 seconds after each operator confirmation and then acquires 30 readings at one-second intervals. It rejects raw readings at or beyond 100 and 65435, separation of 500 counts or less, reversed order, or a capture spread greater than 0.95 estimated pH. It saves both complete captures, their means, operator initials, an identifier and provisional reference status. [E04]

The conversion is:

pH estimate = 4.00 + 3.00 × (raw − R4) / (R7 − R4)

R4 and R7 are the saved mean raw readings. The final archive bench-59daedbc contains stored means 31220.868 and 38102.332. Recomputed means differ by only about 0.0013 counts. Its estimated-pH spreads are 0.06278 and 0.04883, both below the configured 0.95 capture criterion. This describes within-capture variation, not independent accuracy. [E05]

STAGE	LATEST IMPLEMENTATION
Start	NEW_SAMPLE clears the preceding capture, filter, stability and result; duplicate delivery of the same ID does not restart it.
Acquire	One raw reading in each scheduled second 1-20, with a 250 ms acquisition tolerance.
Filter	Three-reading moving average of converted pH values.
Settle	Latest ten unfiltered values have range ≤0.35 pH for a ten-second dwell.
Evaluate	At 21 s, check calibration, completeness, signal range, pH validity, mode-specific response or drift, and settling.
Hold	Freeze a valid completed result; continue status and button handling without new ADC acquisition.

The routine result is the three-reading average at second 20, using readings 18–20, classified before rounding. A steady value alone cannot identify a stuck sensor because two unknown samples may have similar pH. We therefore use a separate known-buffer response check: the operator records the previous sample baseline, moves the probe to a buffer with a known different pH and explicitly starts the response check. If the filtered signal remains within ±0.10 pH of the baseline throughout the specified observation interval, the software reports STUCK_FAILURE. A meaningful response that also meets the settling rule returns PROBE_RESPONSE_DETECTED, not a feeding verdict. Routine samples are not assessed for stuck behaviour. [E03]

The calibration archive has no completion timestamp. Runtime log E08 records an operator-entered completion time of 10 November 2026 at 14:35 UTC for the same calibration ID. The registration is located in records 113–114 of the original E08 JSONL file; these are log line numbers, not report pages. The time is operator supplied rather than an independently timestamped capture. [E05, E08]

Alarms and operator interaction

The browser starts in Ready. During measurement it withholds the main pH and verdict while counting down. A completed result is held; a failed capture displays its technical reason. Raw information, calibration metadata, history and response-check controls are available under More info. [E13]

STATE	CURRENT RING	PROGRAMMED LAPTOP PATTERN
NORMAL	Green	Silent.
ADVISORY	Amber	Three 800 Hz tones, 150 ms each.
CRITICAL	Red	Ten 1200 Hz tones, 150 ms each.
TECHNICAL	Cyan	One 1500 Hz tone, 100 ms.
Measuring	Dim white	Muted.
Physically stopped	OFF	Muted.
Host reconnecting	No confirmed host control of ring	Muted in the latest browser service.

The light-ring and alarm operation was recorded in demonstration videos. With pH above 6.0, the monitor shows Critical, the ring turns red and ten bursts near 1200 Hz support the selected alarm pattern. The audio alarm ceases when the physical ACK/STOP button is pressed. The two recordings are identified as E19 (light and alarm demonstration) and E21 (ACK/STOP demonstration). [E19, E21]

Project evidence location: Team 4 GitHub repository

An on-screen acknowledgement mutes audio for 120 seconds. The physical button generates an ACK event and stops the Pico sample. The Pico repeatedly sends OFF to the ring after stopping. The host removes the current measurement from its main result while retaining the last measurement in the audit. Starting again requires pressing NEW SAMPLE.

The host’s critical latch is a separate memory. It remains until an acknowledgement has occurred and a valid normal reading returns. The current browser panel and ring may nevertheless show a newer advisory sample, or turn off after stopping. This was introduced to avoid displaying an old critical category as the current sample’s result. It also means the ring is not a continuously latched critical indicator. [E03, E13]

After communication loss, the service clears the current sample gate, withholds the host verdict and mutes sound. The browser monitor states “Pico not connected”, a blank pH, paused timer and “Connection lost - reconnect USB”. [E22] It cannot prove the ring’s physical state if the Pico remains powered independently. Closing the browser tab does not stop the service. Restarting the service creates a new session and does not reconstruct critical latch or consent state from historical storage.

Audit model and design alternatives

Each saved event contains received_utc, run_id, operator, sample label, kind and payload. Measurement payloads include sample_id, device elapsed time, pH, raw ADC, calibration ID, validity, state and quality-check information. Device time and host reception time are different clocks and are not interchangeable latency measurements. [E13]

SQLite stores events in one table. A separate JSONL file retains each service run. The service writes and flushes JSONL before inserting and committing SQLite. These writes are not one transaction across both stores. The current snapshots reconcile, but an interrupted write can leave them inconsistent. There is no demonstrated automatic reconciliation or durable replay of the in-memory alarm and recipient state.

The browser shows recent events rather than a complete persistent audit browser. Historical data remain accessible in SQLite and JSONL. An earlier label correction is preserved in a separate sidecar file without altering the original records; it is not an implemented label-editing API. [E12, E13, E15]

CHOICE	RATIONALE AND REMAINING TRADE-OFF
MicroPython	Retained the working Thonny environment and machine ADC interface. The runtime substitution is declared.
Browser and local REST service	Replaced the earlier Tkinter interface and supports the course's browser/service architecture. The backend uses Python's standard HTTP server, rather than FastAPI or Flask.
Laptop audio	Avoided reusing the failed amplifier. It introduces dependence on Windows, the service and laptop speakers.
ntfy Android messages	Provides a demonstrable phone path. It substitutes for the suggested WhatsApp options; specific approval of this substitution is not documented.
Held completed results	Prevents later probe movement from changing a completed sample. A new measurement must be explicitly requested.
Dedicated response challenge	Avoids misclassifying similar unknown samples as a stuck probe. It depends on the operator actually transferring between known buffers.

The revised service offers notifications for completed Normal, Advisory, Critical and Technical results. Technical messages identify why a verdict was withheld. Consent is logged before queueing. The public ntfy topic is accessible to subscribers, so messages contain bench sample information only. [E23]

III. Component, module and interface (Deliverable D3)

Component and module table

MODULE	COMPONENTS AND ROLE
Sensing	Glass probe, BNC lead and pH meter V1.1 board condition the liquid response.
Control	Pico 2 W, Grove shield and MicroPython perform timed acquisition and quality gating.
Local interaction	Sixteen-pixel ring and momentary button provide visual status and physical stop.
Mechanical	Printed case, lid, probe holder and sample-cup position organize the assembly.
Host	Windows laptop, Python service, browser, SQLite and JSONL provide interface, audio and audit.
Remote	ntfy service and an opted-in Android phone provide optional notification.

Interfaces

INTERFACE AND DIRECTION	SIGNAL AND RANGE OR CONSTRAINT
USB → Pico and shield	Nominal 5 V power plus serial communication. Team meter observations were approximately 5.1 V.
Shield supply → sensor + and ring V+	Nominal 5 V supply; common ground. Actual harness continuity is not established solely by code.
Sensor A → GP26 through A0	Nominal ADC domain 0-3.3 V; read_u16 returns 0-65535 counts. Software gate 500-60000 inclusive. Direct wiring has no documented overvoltage protection. [E16]
GP15 → ring DIN	Pixel data at nominal 3.3 V GPIO levels; sixteen RGB pixels. Ring supply is 5 V; level compatibility has not been physically qualified. [E16]
Button → GP14 and GND	Active-low 3.3 V input with internal pull-up; pressed to GND; 30 ms debounce. [E16]
Pico ↔ laptop	Newline-delimited JSON measurements and commands; host opens the port at 115200.
Browser ↔ service	Local HTTP JSON at port 8765; POST requests use a per-process request token.
Service → storage	Append JSONL and commit SQLite event records.
Service → ntfy → phone	Internet-dependent HTTPS messaging; delivery confirmation is a separate operator action.

The hardware uses a Raspberry Pi Pico 2 W connected through a Seeed Studio Grove Shield to the analogue pH board and 16-pixel NeoPixel ring. The final blue enclosure, lid-mounted ring, button and probe support organize the assembly and sample handling. The team reports PLA printing at 15% infill. This is a demonstration enclosure; production mechanical strength has not been established. Splash guards surround the openings to reduce exposure to water or aspirate spills, but no water-resistance rating is claimed.

The speaker is connected to the laptop rather than the Pico amplifier path. The team reports a PAM8302A short circuit during soldering, after which laptop-driven audio was used. The amplifier is excluded from the final working system. Earlier assembly photographs are available in the repository hardware folder; the cited videos document the enclosed final appearance. [E17, E19, E21]

Hardware photographs: GitHub hardware folder

IV. System diagrams (Deliverable D4)

Wiring and control diagrams

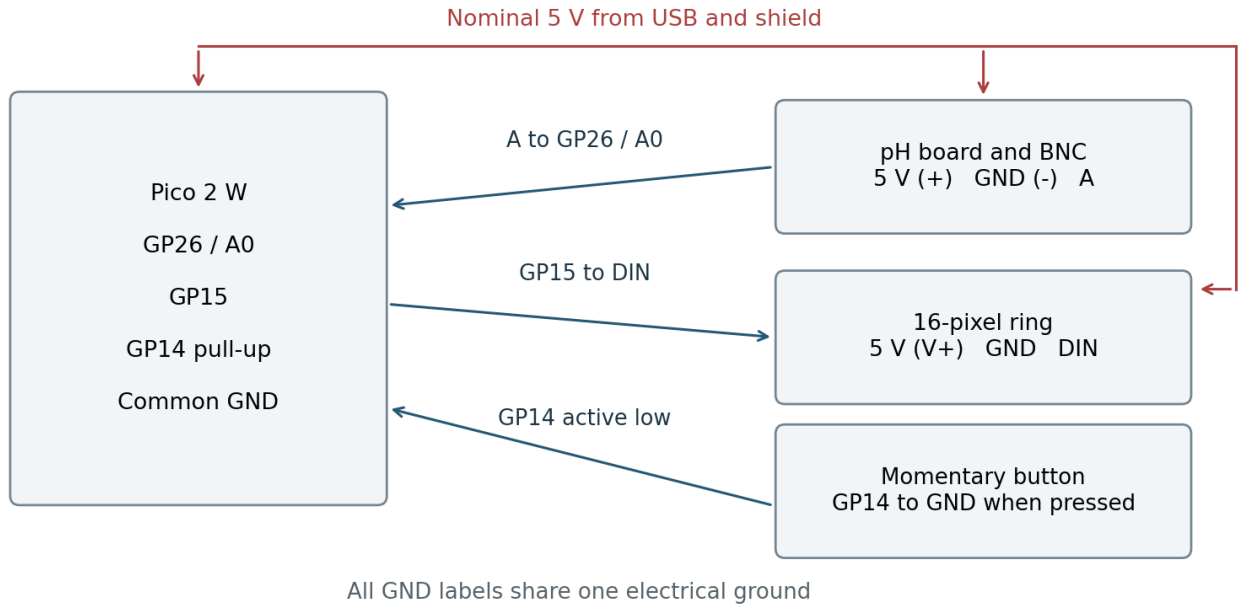
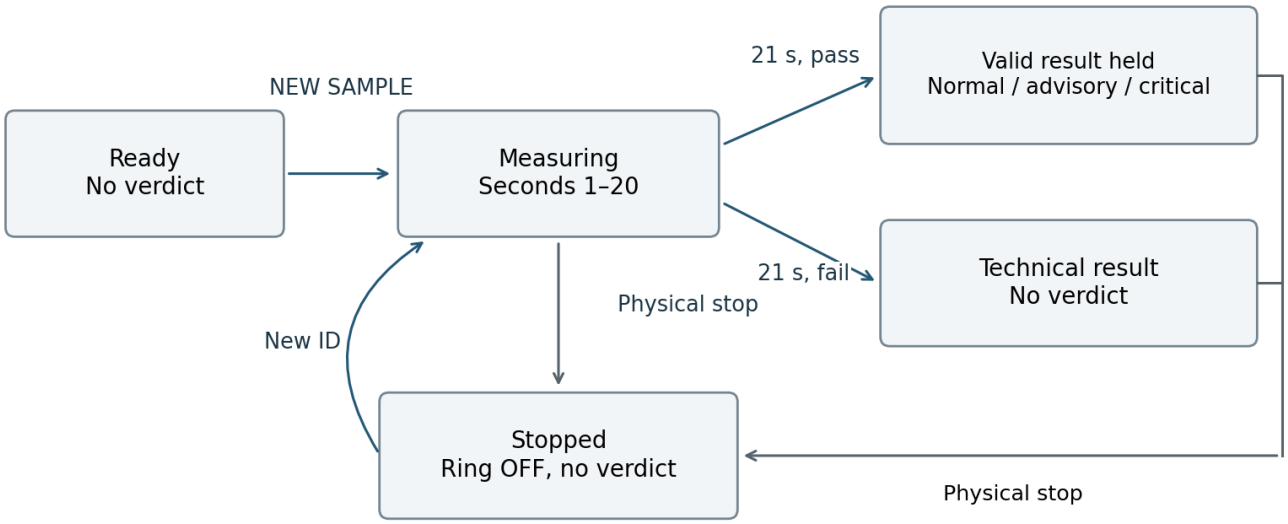


Figure 2. Functional wiring for the documented signal assignments. Grounds are common. The sensor-to-ADC link is shown as documented direct wiring, not as an electrically protected input. Wire colour is not used to infer a pin assignment. [E03, E17]



NEW SAMPLE with a new ID can also restart from either completed outcome.

Host critical memory	Retained after ACK; clears only when a valid normal returns.

Figure 3. Pico sample lifecycle and the separate host critical memory. A completed valid result is held. A technical result withholds a verdict. Physical stop commands OFF; it does not erase the host's critical history. This explains why our alarm system differs from the course. [E03, E13]

Sample to notification sequence

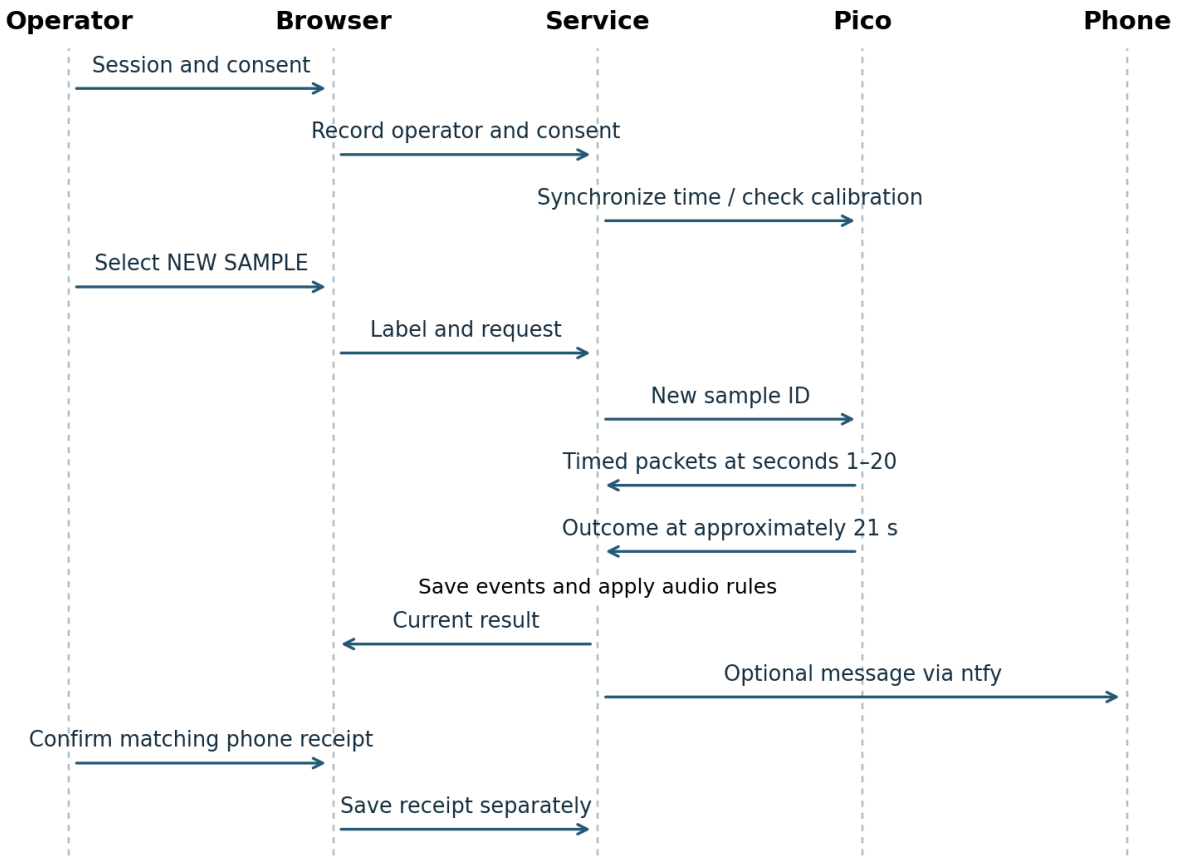


Figure 4. Routine measurement and notification sequence. The decision time starts when the Pico accepts the sample. The notification branch applies to all four completed outcomes in the revised service. [E03, E13, E23]

The service synchronizes laptop UTC with the Pico. If calibration completion has not been registered, the operator supplies the actual completion time once; that control does not perform a calibration. Each NEW SAMPLE creates a fresh identifier. Repeated requests with the same identifier do not restart the Pico timer.

With consent active, the revised service queues one submission per sample ID for an eligible result at or after 21 seconds. Measuring, idle and manually stopped samples do not send. USB loss clears the active sample and has no phone alert. All four outcomes use ntfy priority 4. The phone worker records provider acceptance or failure. The operator records receipt separately. [E23]

Communication interruption clears the host’s current sample gate. Reconnection does not convert buffered or held data into a valid new measurement. The operator must request another sample. This behaviour does not establish recovery from every power, storage or process failure; the service’s in-memory state is not reconstructed after a fresh process start.

V. Implementation and deployment (Deliverable D5)

ITEM	REVIEWED IMPLEMENTATION
Pico firmware	Attached main.py; identical to button_stop_update/main.py; reports sample_response_v10.
Calibration	Attached calibrate_bench_weekly.py and bench-59daedbc.json with all 60 raw readings.
Backend and rules	browser_monitor/service.py, monitor_rules.py and phone_notifications.py.
Browser	browser_monitor/index.html, dashboard.js and styles.css; custom Canvas trace.
Dependency and launch	browser_monitor/requirements.txt specifies pyserial; start_browser_monitor.cmd starts the service on Windows.
Evidence	Original archives/PDF; added meter snapshots, two videos, connection-loss image, source hashes, claim map and trial index.
Printed assembly	Enclosure, fitted ring/button and support visible in the video; manufacturing revision and mechanical performance remain unverified.

Our repository is https://github.com/EXSU-620-TEAM-4-NG-TUBE/ng-tube-prototype. The reviewed versions are sample_response_v10 on the Pico and browser_service_v2_all_results on the host. [E02, E03, E13, E23]

Deployment procedure

1. Use the team's electrically checked bench assembly. Save the supplied MicroPython firmware on the Pico as main.py. Keep the failed amplifier isolated.
2. Run the calibration script in Thonny when recalibration is required. Retain the complete calibration record and use the same sensor, wiring and acquisition configuration for subsequent measurements.
3. Close Thonny before opening the serial monitor. In the Windows Python environment, install browser_monitor/requirements.txt and start start_browser_monitor.cmd. Keep the service window open.
4. Enter operator initials and a session label. Confirm the actual port and loaded calibration ID. Register a missing completion timestamp using the actual calibration time, rather than resetting an old calibration to the current time.
5. Rinse, immerse and keep the probe still. Enter a sample label and select NEW SAMPLE. Observe the outcome after the 21-second acquisition workflow and retain its sample ID.
6. Use on-screen ACK for audio muting or the physical button to stop. After reconnection or stopping, select NEW SAMPLE for another measurement.
7. Restart the Python service after updating both notification files and confirm version browser_service_v2_all_results at /api/health. Record consent and check one matching phone receipt for each completed outcome.

The latest firmware does not consume calibration_pending.json or enforce the older one-use ten-minute session. The calibration script still writes that legacy file; current authorization depends on a usable saved calibration and clock. [E03, E04]

VI. Verification plan (Deliverable D6)

The verification plan separates acceptance criteria from observed outcomes in D7. Some criteria were established during development rather than prospectively. Software injections assess defined logic; physical observations assess the demonstrated assembly. Repeated held packets are not independent trials.

ER	METHOD AND REVIEWED SAMPLE SET	ACCEPTANCE AND REVIEWER
ER1	Recompute 30 readings per calibration buffer; retain nominal-label checks and three added paired meter comparisons from one supplied snapshot run.	Each paired difference must be within ±0.30 pH. Document comparator identity, calibration status, uncertainty and liquid temperature. Measurement reviewer; sign-off pending.
ER2	Reconstruct five v10 captures and 100 scheduled readings; execute virtual clock/ADC timing and missing/late-reading cases.	No early verdict; 21.000-21.250 s outcome; complete timed capture and defined stability gate. Firmware and verification reviewers.
ER3	Execute ten current-source boundary/classification cases and eight invalid/gating cases; compare five v10 outcomes.	Every supplied valid input has the correct unrounded category; invalid or gated input has no verdict. Firmware reviewer.
ER4	Run virtual drift, stuck, signal and incomplete-capture cases; inspect four earlier physical response challenges and retained refusals.	Defined injected failure gives technical/no verdict. Document routine versus challenge scope and physical versus virtual input. Verification reviewer.
ER5	Inject missing time, future time, restart age and expiry boundary; inspect registered completion and current archive.	No age renewal; no new verdict for invalid age. A separate fixed-calibration seven-day reference study is required. Firmware and measurement reviewers.
ER6	Review five v10 stops, historical v9 normal clearing/v3-v8 active stops; inspect both videos for colours, tones and physical stop/mute.	Required colours, tone sequences, physical STOP/mute/OFF and separate critical history must behave as specified. Record physical observations and any untested pattern. Hardware/interface reviewers; sign-off pending.
ER7	Compare three paired stores; inspect interruption records, connection-loss image and five service unit tests.	Corresponding stored fields agree; no interrupted sample becomes a new valid sample; records and corrections remain retrievable. Service reviewer.
ER8	Software: verify every defined technical reason maps to the correct message. Phone: run normal, advisory, critical and one representative technical result on the revised service, matching sample IDs to consent and receipt.	Four matching phone receipts establish category delivery only. Verify no duplicates, pre-result or stopped-sample messages. Claim receipt of every technical subtype only when each has its own matched receipt. Team notification reviewer; sign-off pending.

Team ownership recorded in the notes is firmware by Sophia, Seoyoung, Saada and Bey; dashboard by Sophia; physical design by Bey and Saada; verification and documentation by the full team. These roles identify review responsibility without fabricating a completed sign-off. [E01]

VII. Verification report (Deliverable D7)

The original archives contain 139 JSONL copies and 123 distinct byte streams after hash deduplication. Copies and held records are not additional trials. Added evidence comprises three user-transcribed meter comparisons, two videos, one connection-loss image and a retrospective alarm-record index. These additions do not change the original JSONL counts or store reconciliations. [E15, E18-E22]

DATASET	JSONL FILES	EVENTS	INTERPRETATION
Separate bench_logs.zip	80	9301	Earlier bench development and trials.
Project button_stop_update logs	33	4734	Multiple stop, fixed-window, weekly and response revisions.
Project browser_monitor logs	10	3928	Browser integration, including v8, v9 and v10.
Preserved historical logs	11	1526	Copies of earlier development evidence.
Preserved latest_update logs	5	1150	Selected historical copies and named evidence exports.

Counts above are per location and include duplicate copies between locations. They are not summed to estimate experiment count. The browser set contains 40 NEW_SAMPLE_REQUEST events, 24 FIXED_WINDOW_RESULT events and 3,238 READING records. Its sample index has 41 run/sample associations, including an inherited association without a new request in that run.

Paired storage result

All three databases returned “ok” from SQLite integrity_check. Event comparison normalized payload JSON and matched every stored field, including time, operator, label, kind and payload, while preserving duplicate counts.

STORE PAIR	JSONL	SQLITE	MATCHING EVENTS	DIFFERENCE
Browser	3928	3928	3928	None.
Stop monitor	4734	4734	4734	None.
Earlier bench archive	9301	9300	9300	One JSONL-only RUN_START.

The unmatched event belongs to run 6ad22b889efb41119ec7a7f0856189fc, 2026-11-03 01:21:07.387719 UTC, label “nominal pH 7”. It is a session-start record, not a missing measurement. Its cause is not established; snapshot timing and interrupted persistence are possibilities, not findings.

This gives us more confidence that the current data records are consistent, although we still found one discrepancy in an older dataset. Snapshot equality does not prove atomic writes, disk-failure recovery or restart replay. [E15]

Final calibration model and residuals

Calibration bench-59daedbc is identified by the five final v10 samples. It contains 30 raw readings per nominal buffer, operator SL and reference_integrity marked UNRESOLVED. Figure 5 uses these final capture data. [E05-E07]

QUANTITY	NOMINAL PH 4.00	NOMINAL PH 7.00
Capture count	30	30
Recomputed raw mean	31220.8667	38102.3333
Stored raw mean	31220.868	38102.332
Raw minimum to maximum	31159-31303	38041-38153
Estimated pH range within capture	0.06278	0.04883

The raw separation is 6881.4667 counts using recomputed means, giving approximately 0.000435954 pH per count. The tiny mean differences are consistent with numeric rounding and do not affect the displayed result.

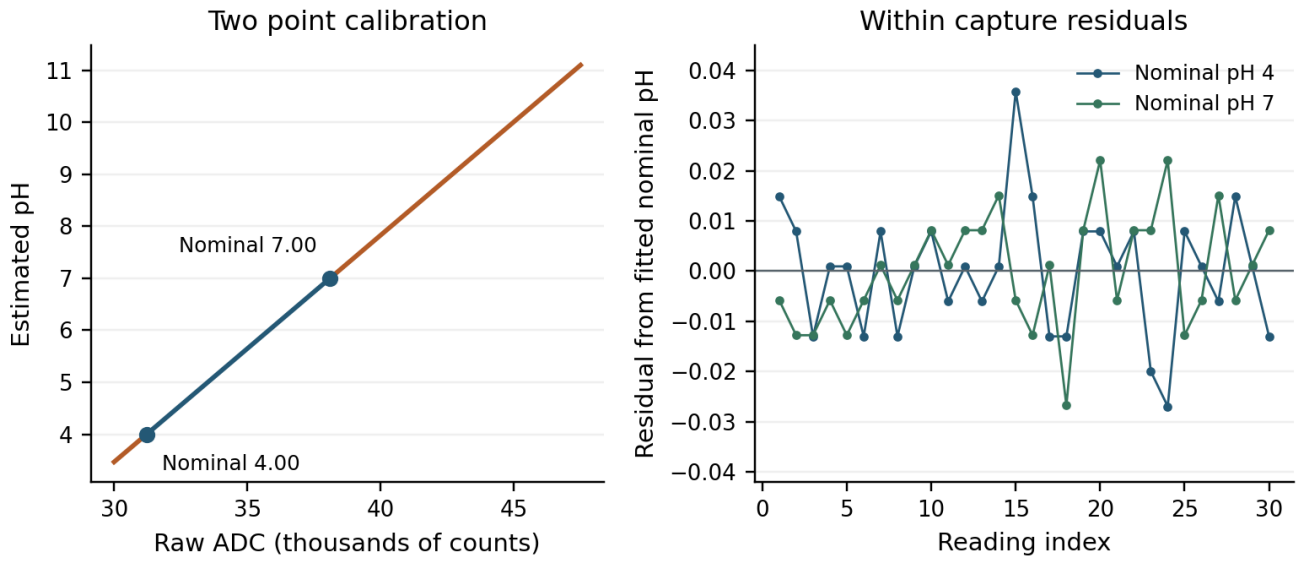


Figure 5. Final two-point model and individual capture residuals about the fitted nominal levels. Blue denotes the fitted 4-7 interval; orange denotes extrapolation. Residuals use recomputed means. All points are from the supplied calibration archive. [E05]

The final captures pass the configured 0.95 spread screen and would also pass a 0.20 screen. This measures within-capture scatter, not independent accuracy: fitting a line through two means reproduces those means by construction. Thirty adjacent observations are not thirty independently prepared standards.

Reference lot, certified uncertainty and actual liquid temperature are not documented in the final record. Earlier notes describe buffer labels and expiry. The added digital-meter readings permit three paired comparisons, but do not establish the meter as traceable ground truth. No R-squared, zero fitting residual or successful save operation is presented as independent validation. [E18]

Reference checks and meter agreement

Independent accuracy requires a reference that was not used to determine the measured result. The following checks retain their actual revision and calibration context. Nominal labels are not independently established true values. [E08, E09, E11]

EVIDENCE AND NOMINAL LIQUID	OBSERVATIONS	ESTIMATE	DEVIATION
Historical T28 return to 7.00	30 within one immersion	7.021 mean	+0.021
Historical T28 held-out 10.01	30 within one immersion	10.360 mean	+0.350
v9 PH4-labelled samples	Two completed samples	4.224; 4.175	+0.224; +0.175
v9 PH7-labelled sample	One completed sample	6.985	−0.015
v9 PH10.01 and PH10.01 A	Two completed samples	10.564; 10.550	+0.554; +0.540
v10 PH10-labelled sample	One completed sample	10.585	+0.575 if nominal 10.01

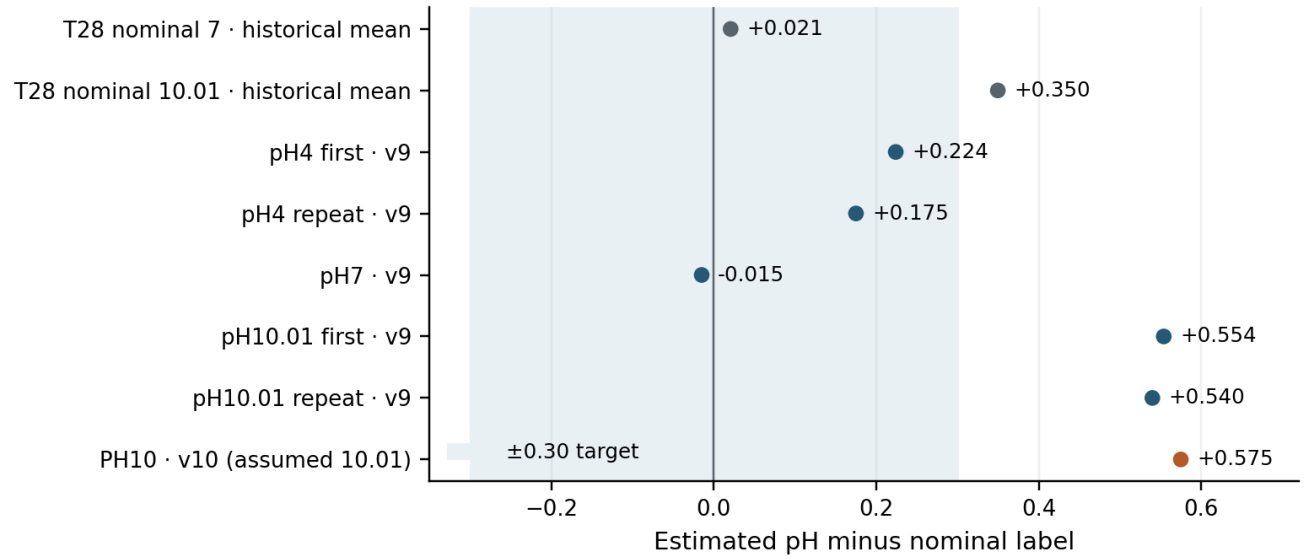


Figure 6. Selected deviations from nominal labels against the ±0.30 target. Different revisions and calibrations are shown separately and are not pooled into an accuracy statistic. The final PH10 comparison assumes the liquid was the nominal 10.01 buffer. [E07-E09, E11]

For historical T28, the independently recomputed 90-120 s windows contain 30 observations each. The pH 10.01 range is 10.280-10.410; the return-to-7 range is 6.972-7.112. The high-pH point was not used to fit that calibration. [E11 lines 116-145 and 381-410]

Several historical differences from nominal buffer labels exceed ±0.30. The new high-pH comparison instead uses the meter's observed 10.41 and the prototype's 10.641, a difference of +0.231. This changes the comparison reference; it does not prove that the earlier discrepancy was corrected or identify whether the buffer, meter or prototype caused it. Both sets are retained separately.

Paired digital meter comparisons

The team supplied three digital-meter readings paired with device snapshots from run 20fda0d966d2425e a3a0105a0911b8c1. All identify calibration bench-59daedbc and firmware sample_response_v10. E18 preserves selected fields as a user-supplied transcription; the original JSONL was not used for this snapshot analysis. These checks are separate from the nominal-label comparisons. [E18]

SAMPLE ID PREFIX	METER PH	PROTOTYPE PH	DIFFERENCE	±0.30 CHECK
5b75e456	3.94	4.022	+0.082	PASS
0e142cf6	6.91	7.015	+0.105	PASS
235f4850	10.41	10.641	+0.231	PASS

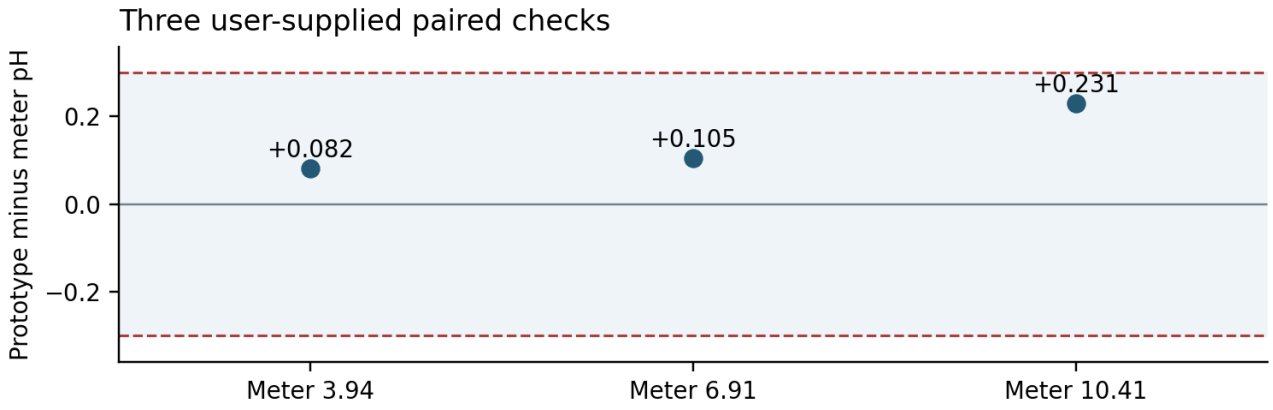


Figure 7. Three paired prototype-meter differences against the ±0.30 numerical target. No uncertainty bars are shown because comparator uncertainty was not supplied. The plot establishes selected agreement, not reference-traceable accuracy over a continuous range. [E18]

The first value is the completed current result: VALID, NORMAL, sample_complete and result_held true. The other snapshots were taken after stopping; their current pH and verdict are null by design. Their retained last_measurement entries contain the valid completed results 7.015 and 10.641. Those saved values, rather than the null current fields, are used above. Packet values have three decimal places and are not represented as full internal precision.

All three absolute differences meet ±0.30. Maximum absolute difference is 0.231 pH and mean absolute difference is 0.139 pH for these three pairs only. All differences are positive, suggesting a possible systematic offset to investigate; three observations do not establish bias. There is one observation at each level, not a repeatability study. No intermediate or near-threshold accuracy is established.

The prototype agreed with the comparison meter within ±0.30 pH at three tested points. Meter model, calibration status, uncertainty and liquid temperature were not recorded in the supplied comparison data. Reference uncertainty, repeatability and accuracy near the decision thresholds remain unverified. The calibration_valid flag indicates software eligibility, not physical accuracy.

Final workflow timing and held results

Five completed routine samples in two browser runs identify firmware sample_response_v10 and calibration bench-59daedbc. The table contains one decision per sample, rather than counting repeated held packets as repeated tests. [E06, E07]

SAMPLE ID PREFIX	FINAL PH	CATEGORY	DEVICE TIME S	RANGE PH	NET CHANGE PH
3609e8b9	7.006	Critical	21.004	0.056	+0.014
9472776b	6.029	Critical	21.008	0.042	+0.007
d66e2bff	5.950	Advisory	21.008	0.042	+0.030
f4ce4d6e	5.908	Advisory	21.009	0.049	−0.014
d9127ee4	10.585	Critical	21.007	0.105	+0.012

All 100 scheduled readings are present. Recorded acquisition lateness ranges from 0 to 10 ms, below the configured 250 ms tolerance. Every final capture reports complete_timed_capture true. Recomputing the mean of converted raw readings 18-20 reproduces all five final pH values within packet rounding. No pre-21-second verdict occurs in these reviewed captures. [E15]

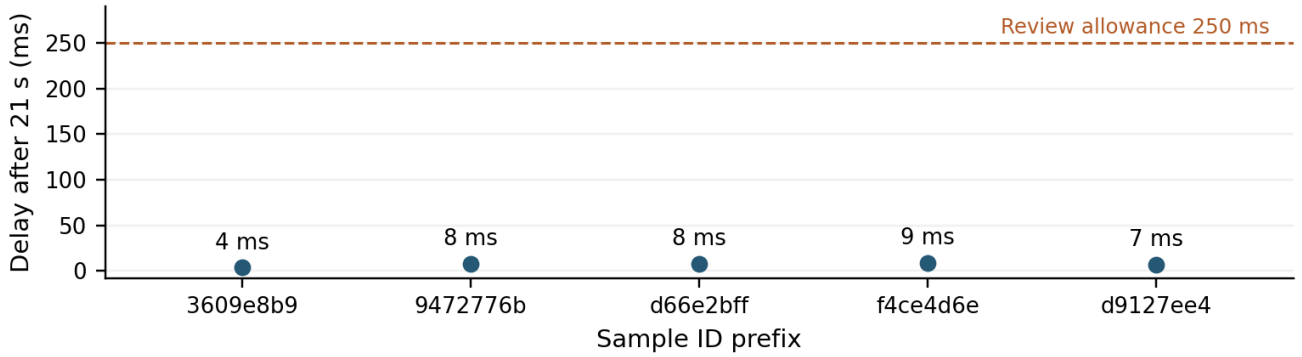


Figure 8. Device publication delay beyond 21 seconds for the five completed v10 samples. The observed delay is 4-9 ms; the review acceptance allowance is 250 ms. This is software publication timing, not physical electrode response or sound latency.

Host request-to-result-event intervals are 21.051-21.090 seconds. Those intervals include serial command and event handling; the physical start of immersion is not synchronized to them. ER2 passes for these reviewed timing cases and the virtual incomplete/late-capture tests.

Each sample has a physical stop event after completion. Subsequent records command OFF and withhold the current verdict. The final PH10 run contains 285 OFF heartbeats after its one stop, not 285 separate stop trials. A 60-second virtual run changes input after completion and confirms the held result and absence of further ADC acquisition. A separate run verifies that NEW SAMPLE resumes acquisition. Active-acquisition stopping passes virtually. Earlier physical v3/v8 stops are indexed separately; a final-v10 physical active-acquisition stop is not identified. [E20]

These five fully archived v10 captures contain no normal result. The added E18 snapshot supplies a distinct valid v10 NORMAL result of 4.022, and the video shows a green 4.0 display. Neither addition is represented as a sixth fully reconstructed 20-reading capture. [E18, E19]

Fault rejection and response checks

The latest code contains explicit fault logic; the PDF's "need to make the code" note is superseded for the defined cases. Its claim that all drifting conditions are detected is not supported. Drift is assessed in routine mode; known-buffer response mode permits the expected pH change and uses the separate response and settling checks. [E01, E03]

TEST INPUT OR CONDITION	CURRENT SOFTWARE EVIDENCE
Routine net change >0.40 pH	DRIFT_FAILURE and no verdict; equality and both directions are exercised.
Confirmed buffer change with unchanged filtered signal	STUCK_FAILURE and no verdict; inclusive ±0.10 boundary is exercised.
Similar consecutive unknown samples	No stuck inference; the same pH may be legitimate.
Early meaningful response in challenge mode	PROBE_RESPONSE_DETECTED; not a feeding verdict.
Late response or excessive range	UNABLE_TO_SETTLE and no verdict.
Raw outside 500-60000	SIGNAL_OUT_OF_RANGE at decision; inclusivity checked at both limits.
Missing or late timed reading	MISSING_TIMED_READINGS and no verdict.

Four v9 known-buffer challenges are identifiable: two 4→7 challenges and one 10.01→4 challenge report response detected; a 7→10 challenge is rejected by the older signal range. Thus three have valid response outcomes and one is refused. These are tests of response under stated transfers, not three trials with a genuinely stuck electrode. [E08, E09]

The labels "PH4 STUCK TEST" in two routine v9 trials do not establish a stuck challenge: their requests have known_buffer_changed false, and the saved results explicitly say NOT_ASSESSED_ROUTINE_SAMPLE. This prevents a misleading test name from being counted as demonstrated fault detection. [E08]

The PDF's small before/after differences, such as approximately 4.20 to 4.16, do not exceed the configured 0.40 drift threshold. They cannot demonstrate rejection of a deliberately drifting input. A correctly responding physical probe also cannot establish the sensitivity of the stuck detector.

Defined fault rejection is demonstrated under demo conditions for the virtual inputs and selected buffer challenges. Routine unknown-sample stuck detection is not implemented. Physical fault sensitivity and specificity are not estimable: no independent labelled set of genuine faulty and healthy electrodes, with TP, FN, FP and TN counts, is supplied. Passing the software tests does not tell us how reliably the prototype would detect faults in a real physical sensor.

Calibration age and software verification

The weekly gate is calculated from saved calibration completion and synchronized laptop UTC. It returns no valid age when time is missing, completion lies in the future, coefficients fail validation or seven days have elapsed. Restarting on day three leaves four days; it does not create a fresh week. The saved calibration timestamp cannot be registered again to renew an existing record. [E03]

Runtime E08 records registration for bench-59daedbc and the five v10 results carry the same completion epoch. The calibration archive's null timestamp shows why archive identity and operational state must both be retained. The timestamp's factual accuracy still depends on the operator and laptop clock.

No fixed-calibration seven-day physical comparison is identified. Logs collected on several dates under different calibration IDs do not constitute that study. ER5 calibration-age gating is demonstrated under demo conditions through the selected software cases. Physical accuracy throughout seven days remains unverified.

Software verification by version

Ten supplied check scripts passed on copies of the reviewed code. The earlier browser-service suite contained five tests. After the all-results notification update, seven tests passed across test_service and test_notification_messages for browser_service_v2_all_results. This later count includes the existing tests and is not seven additional tests. Simulated serial inputs and mocked delivery were used; no phone messages were sent by these checks. [E14, E23]

CHECK	SCOPE
check_firmware_loop	Actual firmware control loop, publication, held result, new sample and repeated OFF after stop.
check_fixed21	Decision block, six drift readings, missing/late capture and calibration precedence.
check_weekly	Expiry equality, missing clock, future timestamp and restart age.
check_signal_limits	Actual current ADC constants and inclusive boundaries.
check_stuck and check_drift	Defined thresholds, duration, known-change scope and selected exceptions.
check_logic and check_countdown	Stop idempotence, latch, duplicate prevention and countdown; includes older desktop control helpers.
check_simple_monitor and check_phone_notifications	Earlier desktop loop and mocked message acceptance/failure; supplementary, not final-browser hardware evidence.

The old check_deadline script targets an earlier 20-second implementation and was excluded. check_fixed21 supplies its own older ADC maximum, so it is used for decision/timing evidence, not proof of the final signal limit. Separate current-source classification checks passed ten numerical cases and eight invalid/gating cases. All results are software-only evidence.

Alarm notification and interruption evidence

Five v10 physical ACK/STOP actions are independently recoverable from E06/E07. They are followed by 387 stopped/OFF heartbeats with no current verdict. ACK-to-STOP device event intervals are 1-2 ms; these are software intervals, not physical press-to-light or press-to-sound latency. The videos add direct physical observations, including audio cessation during ACK, in the following section. [E20, E21]

Retrospective review also identifies a complete earlier browser sequence: valid critical 7.059, physical ACK/STOP, retained critical history, then a routine valid normal 4.224 and cleared history. This is v9 evidence, not a final-v10 repetition. The routine request has known_buffer_changed false despite its “STUCK TEST” label. [E08 lines 407, 418-421, 644, 669-672]

Earlier v3 physical stops at 12.272 and 8.667 seconds remain stopped past their original 20-second deadlines, without a new current verdict. A v8 stop at 4.501 seconds has only about seven seconds of follow-up. These demonstrate historical active-stop behaviour; current firmware is supported separately by virtual tests. Final-v10 history also remains latched through ACK, OFF and later advisory results. [E20]

The chosen design keeps current indication separate from critical history. A physical stop turns the ring off; the host latch remains until acknowledged and a valid normal returns. Fresh service startup does not restore it. The disconnected-state image shows no current pH, a paused timer and an explicit reconnect message. It does not record recovery or detection latency. Reconnection is muted; an older USB-loss chirp is not final-browser behaviour. [E22]

Phone and interruption evidence

Historical browser logs contain 13 consent events, 24 queued/accepted messages and two manual receipt confirmations. Revised software tests cover category eligibility, technical reason text, consent, timing and duplicate suppression. Live acceptance requires normal, advisory, critical and one representative technical receipt with matching sample IDs. That verifies category delivery, not every technical subtype; each subtype needs its own receipt if that stronger claim is made. [E07, E09, E19, E23]

FINAL PH10 DELIVERY MILESTONE	UTC ON 10 NOVEMBER 2026
Recipient consent	20:50:00.546578
Sample request	20:50:32.362650
Fixed-window result received	20:50:53.413479
Matching phone receipt manually confirmed	20:51:18.441864

The final confirmation links sample d9127ee4… to message t0p7nGtG1Z0p; the second links v9 sample 997b090e… to 6JNqJDP32EwX. Both are MANUAL_PHONE_OBSERVATION. The 25.03-second result-to-confirmation interval includes human action and is not arrival latency. [E07 line 95; E09 line 301]

The logs retain 47 link errors, 19 interruptions and 72 parse errors; repeated events are not independent failures. Current store equality supports snapshot consistency, while process replay and interrupted-write recovery remain incomplete. [E15]

Physical and connection demonstrations

The recordings show a green ring with a normal result near pH 4.0, a red ring with a critical result near pH 7.0, and physical ACK/STOP followed by ring OFF and audio cessation. The connection-loss image shows “Pico not connected”, a blank pH and a paused timer. These observations demonstrate the displayed functions; they do not establish every fault or recovery condition. [E19, E21, E22]

Recordings and assembly evidence are indexed for the Team 4 repository (https://github.com/EXSU-620-TEAM-4-NG-TUBE/ng-tube-prototype/tree/main)

The repository currently contains logs and earlier assembly photos.

Requirement dispositions

Status key: PASS — TESTED SCOPE means the stated reviewed cases meet their criteria; it is not a universal performance claim. PARTIAL means some functions are supported but the full requirement is not established. The evidence column defines the scope of each status.

REQUIREMENT	DEMONSTRATION DISPOSITION	EVIDENCE SUPPORTED SCOPE AND REMAINING UNCERTAINTY
ER1 Accuracy and range	PARTIAL	Three meter comparisons meet ±0.30: +0.082, +0.105, +0.231. Reference calibration/uncertainty, temperature, repeatability and continuous-range accuracy remain unresolved. Historical nominal-label differences retained.
ER2 Timing and settling	PASS — TESTED SCOPE	Five v10 outcomes at 21.004-21.009 s; 100 scheduled readings and selected software withholding cases pass. Universal electrode response time is not established.
ER3 Classification	PASS — TESTED SCOPE	Boundary tests, invalid-input tests and recorded outcomes support correct classification before rounding. Physical measurement accuracy near the thresholds remains an ER1 issue.
ER4 Fault rejection	PARTIAL	Defined virtual faults and selected buffer challenges support refusal/response logic. Genuine physical fault coverage and routine unknown-sample stuck detection are not established.
ER5 Weekly calibration	PARTIAL	Selected age-gating software cases pass. Seven-day physical accuracy with unchanged coefficients is not demonstrated.
ER6 Indication and stop	PARTIAL	Videos directly support green/red indications, critical tones and physical ACK stopping/muting with ring OFF. Logs support critical-history retention.
ER7 Identity audit recovery	PASS — TESTED SCOPE	Current JSONL/SQLite pairs match, sample identities and corrections are retained, and interruption handling supports withholding the current verdict. The image confirms the disconnected display. Broader storage-failure and process-restart recovery remain unverified.
ER8 Consent and notification	PARTIAL	Revised software checks cover category eligibility and reason-specific messages. Two historical matched receipts establish earlier delivery. Obtain four revised category receipts; separately retain each technical subtype receipt if claiming all subtypes were received.

The current STOP/OFF design is retained. Current indication follows the current sample; critical history is retained separately on the host. This remains a declared difference from continuously latched red indication.

Remaining validation work

ER1 needs documented comparator calibration/uncertainty, temperature and repeated independent checks across the claimed range and decision boundaries. ER4 needs a labelled physical fault set. ER5 needs an actual fixed-calibration seven-day study. ER6 needs the unfilmed physical patterns and quantified timing for any latency claim. ER7 needs controlled storage/process interruptions and documented restart recovery. These are remaining activities, not invented test results.

VIII. Failure modes and mitigation (Deliverable D8)

ID AND FAILURE	S O D	RPN	IMPLEMENTED MITIGATION AND REMAINING EVIDENCE
FM01 Electrode damage or disconnection	5 1 2	10	Signal gate and response challenge; not every physical fault is detectable.
FM02 USB or communication failure	4 1 1	4	Host withholds result on link loss; no independent audio during outage.
FM03 Liquid exposure	4 2 1	8	PLA enclosure and splash guards; ingress resistance not tested.
FM04 Unstable placement or damage	3 2 1	6	Printed holder and secured assembly; no quantified force or drop test.
FM05 Invalid or expired calibration	5 2 1	10	Coefficient and age checks; reference accuracy and weekly stability unresolved.
FM06 Undetected drift	5 1 1	5	First/last mean and settling checks; selected drift patterns only.
FM07 Stuck or unresponsive probe	5 2 2	20	Known-buffer response check. Routine unknown-sample stuck detection is not established.
FM08 Category switching near boundaries	3 3 1	9	Unrounded classification and held result; repeat samples may straddle thresholds.
FM09 Premature settling	5 1 1	5	Range/dwell checks and 21 s decision; universal electrode settling not established.
FM10 Incorrect or incomplete audit	3 1 1	3	Unique IDs and dual storage; non-atomic writes remain.
FM11 Insufficient sample volume	4 3 1	12	Adequate tip immersion instructed; minimum volume and automatic detection unverified.
FM12 Probe movement	4 2 1	8	Holder and still measurement; movement tolerance unmeasured.
FM13 Inadequate rinsing and carry-over	4 3 3	36	Rinsing required; controlled carry-over study absent.
FM14 ACK fails to stop as intended	2 1 1	2	Button/OFF and audio cessation shown; outage behaviour unverified.
FM15 Browser or notification failure	4 2 1	8	Consent, logging and receipt checks; service and internet dependencies remain.
FM16 Power interruption or restart	5 2 1	10	Time sync and fresh-sample gate; alarm history/consent not restored after restart.

The following scores are qualitative post-mitigation judgments for the implemented bench design, not measured probabilities. Higher scores indicate greater severity S, occurrence O or difficulty of detection D; RPN = S × O × D. [E01]

Scoring anchors for this review: S1 negligible effect, S2 minor disruption, S3 repeat test or workflow delay, S4 lost function or misleading information, S5 potentially consequential incorrect decision. O1 considered unlikely, O2 occasional, O3 plausible recurring, O4 frequent, O5 very frequent; these are judgments without measured rates. D1 readily apparent or expected to be detected, D2 usually detectable with a defined check, D3 detection depends on operator action, D4 difficult to detect, D5 no reliable detection. These anchors clarify interpretation; they do not retroactively validate the retained scores.

The team reports that the unstable readings were resolved by replacing the Pico, rather than changing the cables, and attributes the fault to the original Pico cable connection point. Replacement success is a team observation; the precise contact defect was not independently isolated. The direct sensor-to-ADC path has no documented overvoltage protection, and software signal rejection does not provide electrical protection.

FM13 has the highest retained RPN, but this is not a validated ranking of clinical risk. For FM07, routine unknown-sample stuck detection is not established

The prototype requires a stable surface, adequate immersion, a functioning laptop and a deliberate new-sample action. Probe transfer and rinsing are operator-dependent. The evidence does not demonstrate a minimum 3-5 mL volume, liquid-ingress resistance, cleaning compatibility, sterilization, electrical certification or nurse usability.

The words “safe to feed” and “not gastric” are course demonstration labels. A pH-only bench classification cannot establish actual tube location.

IX. Traceability and limitations (Deliverable D9)

Clinical requirement	Corresponding engineering requirements	Evidence	Status
CR-1	**ER1** — pH accuracy, range, and display. **ER2** — acquisition timing and settling.	The dashboard displayed pH to one decimal place. Five reviewed v10 samples returned outcomes at 21.004–21.009 s. Three pocket-meter comparisons were within ±0.30 pH, with differences of +0.082, +0.105, and +0.231.	**Met**
CR-2.	**ER3** — classification and verdict withholding. **ER6** — alarm indications and STOP behavior. **ER8** — optional phone notifications.	Software checks exercised both classification boundaries. Bench records show all three verdict categories. Video evidence shows selected green and red indications, critical tones, and physical STOP. Revised phone notifications passed simulated checks for all four completed outcomes.	**Met**
CR-3.	**ER2** — settling and timing. **ER4** — fault rejection and response checks. **ER5** — calibration age. **ER6** — technical indication. **ER7** — interruption handling.	Software checks exercised expiry, drift, missing readings, signal limits, and the dedicated known-buffer response check. Connection-loss evidence shows the dashboard withholding the current pH and verdict.	**Met**
CR-4.	**ER5** — calibration identity and age. **ER7** — sample attribution, audit storage, and recovery. **ER8** — notification consent and receipt records.	Two reviewed SQLite snapshots reconciled with their JSONL event logs. The records link sample IDs, operators, results, and calibration context. Two historical phone receipts were recorded separately from provider acceptance.	**Met**.

The bench prototype demonstrated measurement, selected alarm and stop behaviours, and audit retrieval. The revised service adds reason-specific phone messages for completed technical results and extends notification eligibility to normal and advisory results. Four-outcome phone verification remains open.

The prototype agreed with the comparison meter within ±0.30 pH at three tested points. Videos support selected LED and stop functions; the connection-loss image shows the browser withholding pH. The weekly interval and physical fault-detection limits are summarized in the requirement dispositions.

Attribution

- firmware: Sophia, Seoyoung
- dashboard: Sophia
- physical design: Bey and Saada.
- Verification and documentation: all members
- Github repository: Kayla and Sophia
- Final report: all members

Sources and reproducibility

Evidence references identify project records: E13 is the browser monitor software; E19 is the light/alarm video; E21 is the physical ACK/STOP video; E22 is the connection-loss image. The complete E01–E23 reference index accompanies this report for repository upload.

Repository: EXSU 620 Team 4 NG tube prototype (https://github.com/EXSU-620-TEAM-4-NG-TUBE/ng-tube-prototype/tree/main)